TECHNICAL EVIDENCE / VALIDATION

Technical Evaluation and Limitations

JUNIPER VALE | 14 OCT 2026 | REVISION 1

This report records an actual local validation run against the synthetic transformation engine. It separates executed checks from architecture ambitions and unresolved limitations.

Test execution

CHECK	RESULT	EVIDENCE
Automated pytest checks	31 passed	results/pytest.txt
Synthetic work orders	3,451 created	Fixed seed 20261014
Synthetic branches / technicians	8 / 60	Relational validation in check_data
Replay test cases	20 executed	18 category matches, 2 mismatches
True model-based AI calls	0	No model credentials or LLM SDK
Public production API	Not deployed	Local FastAPI only

Replay evaluation: important distinction

The 18/20 category match (90%) belongs exclusively to a handcrafted deterministic pattern replay over 20 labeled synthetic cases. It is not LLM accuracy and is not statistically valid predictive performance. Every input, expected category, actual output and latency is retained in results/replay-evaluation.json. No unsupported accuracy, prompt-injection robustness or external benchmark is claimed.

Controls actually implemented

• Fixed-seed relational validation, unique keys, invoiced-completion references and branch-to-company accounting reconciliation.

• Exact subset assignment checks technician qualifications, shift duration, time windows, inventory availability and appointment overlap.

• Program dependency cycle detection, day-slot technical resource constraints, CRM migration release gate and repeatable Monte Carlo runs.

• Schema-bearing service interpretations, role-limited state transitions, manager approval before synthetic writes, optimistic version checks, duplicate ID rejection and SQLite audit logs.

- Incident note instructions are not executed; suspicious strings route to review-only mode.

Scope and unresolved gaps

The application is a local proof-of-concept. GitHub Pages can host the browser replay, but not the Python API. The replay is derived from committed calculations rather than a live authoritative API.

Boundaries requiring further engineering

• The exact subset solver is not OR-Tools and does not model sequential travel arcs, dynamic dispatch or full 300-appointment VRPTW.

• The AI component is deterministic replay. Real model inference, structured model schema validation and independent benchmarks remain unimplemented.

• Local X-Role HTTP header is only a demo permission control, not identity verification. Do not expose writable endpoints publicly as deployed auth.

• No production secrets, multi-tenant isolation, immutable enterprise audit store, rate limiting, hosted observability or CI pipeline have been configured.

• Synthetic unit economics are not reconciled to real customers, working-capital flows, tax expense or generally accepted full-cost accounting.

• Monte Carlo duration uncertainties are independently sampled; real correlations, learning curves and distribution calibration are not supported.

Reproduction

Install from requirements.txt, execute `python -m lab.generate`, `python -m lab.evaluate`, `pytest -q`, and launch `uvicorn lab.api:app`. Documentation includes the three integrated scenario names and example API calls.

Evaluation conclusion

The project provides an inspectable linked business model, a real constrained subset search, an interdependent program scheduler, a deterministic approval workflow and reproducible test evidence. It is an advanced independent technical case study, not a live enterprise platform or validated AI product.

Evidence provenance

Local execution: 2026-10-14. Test runner: pytest. Corpus: fixed synthetic seed. All historical Juniper Vale portfolio artifacts remain unchanged.